TOM - Muon Tomography

Signature Sheet

Student's Name _____ Partner's Name _____

Suggested reading to start with (see the end of the manual for more):

1. 111B TOM manual
2. 111B MUO manual (if you did not do MUO, you should read through this manual for more context and consider talking through the additional prelab questions with a GSI)
3. Alvarez et al., Search for Hidden Chambers in the Pyramids
4. *Applications of Cosmic Ray Muons*

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. If you are interested in going to a building other than Evans Hall, at least two days before the start of the lab, check in with instructors about whether access needs to be granted to you for any of the spaces you will be taking the detector cart to.
2. What is a muon? How and where are the muons in this experiment produced? Introduce different types of muon decay. **If you have done MUO, you may skip this question.**
3. How do we capture the muon and convert the decay process into an electrical signal so that we can do statistics? **If you have done MUO, you may skip this question.**
4. What do we call a 'coincidence' and a 'stoppage' in this lab? What is the difference, and when does it matter?
5. What is muon tomography or muography? How can it be used to 'map' or 'image' a building or object? What sorts of algorithms are used in tomographic imaging, and in what other contexts is tomography used?
6. One important aspect of this experiment is setting the threshold for considering a pair of pulses a coincidence. Given two pulses with heights y_1 and y_2, and peak times t_1 and t_2, what are possible heights for the product of the two pulses? Assume the pulses are Gaussian. (Hint: you should be showing a plot; consider Δy and Δt.) What does this tell you about setting the trigger level for coincidences?
7. Given the geometry of the two detectors "Bra" (1'×1'×1') and "Ket" (4"×1'×1'), calculate the number of cosmic rays per minute which enter each detector.
8. At what rate will random coincidences occur given some trigger level and the count rates you calculated in the previous question? (Hint: you may want to reference EAX.)
9. Assume a simplified building where floors are made of concrete and air. Using reasonable estimates, how does count rate depend on which story of a building you are on? Given two measurements on two different floors, how long of a measurement do you need to do for the difference in count rate that you measure to be statistically significant, assuming you are only looking at coincidences, not stoppages?

10. Note that the number of muons that stop in the detector is dependent only on the mass of the detector (see Figure 6 of Rossi, linked in the references in the TOM manual). Assume the density of the mineral oil is 0.8 g/cm^3 and that we are looking for stoppages in 'Bra' (the larger detector). If you are now looking for stoppages instead of just coincidences, how does this change your answer to the previous question?
11. Say you are, like Luis Alvarez, looking at the Pyramids of Giza. What angular resolution do you need to detect a chamber of a given length scale if your detectors are a given distance away with some separation between them? What is the smallest 'chamber' we can detect at a given distance with our detectors?
12. If instead of large scale variations (e.g. chambers in geological or archaeological structures) you were attempting to map smaller scale variations (e.g. under or overdensities in a shipping container) what changes might you make to your apparatus?
13. What must you do whenever you take the cart out of the lab?

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. By day 3, you should have seen a difference in count rate based on altitude. Show a GSI/instructor your plot of count rate vs. altitude for several floors of Evans Hall (or approved alternative location), with appropriate fit line and error bars. Is it consistent with expectations?

Staff Signature _____ Date _____

Completed by day 3 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. <u>Pulse thresholds</u>

   Staff Signature _____

2. <u>Coincidence detection threshold</u>

   Staff Signature _____

3. Material absorption coefficients

Staff Signature _____

4. Coincidences vs. stoppages

Staff Signature _____

5. Angular dependence plan

Staff Signature _____